Anchor Paper RAG: Complete Evaluation Results

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MLOps Course Project, Milestone 1

1 Summary

This document lists all 57 MLflow runs from the evaluation: 32 runs that chose settings, and 25 runs on held-back questions (5 setups, each with 4 per-anchor child runs).

- **Served setup (D):** hybrid search through Pinecone with `gemini-3.5-flash-lite`, with faithfulness 0.978 and 5% refusals.
- **Top score (C):** keyword-only retrieval with the same generator, with faithfulness 0.985 and 10% refusals.
- **Earlier project's setup (B):** faithfulness 0.928 and 18% refusals, but higher relevancy when it answers (0.755 against 0.702 for D).
- **Estimated evaluation spend:** $0.47, because repeated settings reused saved answers.

2 How to read the tables

Color key: chosen or served row (also bold) | top score | faithfulness 0.97 or higher | 0.94 to 0.97 | below 0.94 | refusals 20% or higher

- **Faithful.:** share of an answer's claims supported by its retrieved passages. This is the only score used to choose settings.
- **Relev.:** how closely questions generated from the answer match the question asked. **Rel. if ans.** leaves out refusals, which can score well by repeating the question's words.
- **Right paper:** share of a question's source papers found among the retrieved passages.
- **Refusals:** answers using a known refusal phrase, or with no factual claims. **Claims made:** answers with a faithfulness score. **Judge fails:** judge replies that could not be read.
- **Retrieve, Generate, Total:** average seconds per question. **Cost:** estimated dollars for the run's questions, counted even when saved answers were reused.
- **Winner rule:** highest faithfulness. Within 0.01, higher relevancy wins, then lower latency. A run cannot win if it refuses over 34% of questions or makes claims on under 80%.
- **Questions:** each stage used 15 setting-selection questions on one anchor. The final test used 15 held-back questions per anchor, or 60 in total.

Starting setup and what each stage chose

Stage	Anchor	Setting	Started as	Chosen
1	Nested Learning	Judge	gpt-4o-mini	**gpt-4o-mini-rubric**
1	Nested Learning	Chunking	recursive 400/80	**recursive 800/120**
1	Nested Learning	Embeddings	text-embedding-3-small	**text-embedding-3-small**
2	Titans	Retrieval	hybrid + rerank	**keyword**
2	Titans	Passages used	5	**8**
3	TTT RNNs	Prompt	P0 original	**P1 strict citations**
4	Hierarchical Reasoning	Model	gpt-4o-mini	**gpt-4o-mini**
4	Hierarchical Reasoning	Temperature	0.1	**0.0**

Each stage kept the winners from earlier stages, so later stages tested on an already improved setup.

3 Stage results

Stage 1: Judges on Nested Learning

Judge	On real passages	On unrelated passages	Gap	Relevancy	Judge fails	Cost ($)
gpt-4o-mini	0.894	0.083	0.812	0.794	0	0.0091
gpt-4o-mini-rubric (chosen)	**0.921**	**0.041**	**0.880**	**0.789**	**0**	**0.0093**
gpt-5-nano	0.864	0.065	0.800	0.771	0	0.0058
gemini-3.5-flash-lite	0.782	0.023	0.759	0.812	0	0.0024
gpt-oss-20b	0.964	0.117	0.847	0.772	0	0.0024

The same answers scored from 0.782 to 0.964, so the judge with the largest gap won, not the one with the highest score.

Stage 1: Chunking on Nested Learning

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
fixed-512-50	0.891	0.777	0.759	0.90	20%	80%	0	1.75	2.20	3.95	0.0071
recursive-400-80	0.921	0.789	0.780	0.93	13%	87%	0	1.53	2.04	3.57	0.0093
recursive-800-120 (chosen)	**0.950**	**0.784**	**0.772**	**0.93**	**13%**	**87%**	**0**	**1.76**	**1.99**	**3.76**	**0.0090**
semantic-p70	0.921	0.793	0.780	0.93	13%	87%	0	2.87	2.74	5.62	0.0111

Larger recursive chunks scored highest at 0.950, and semantic chunking tied with recursive 400/80 at about 0.921.

Stage 1: Embeddings on Nested Learning

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
text-embedding-3-small (chosen)	**0.950**	**0.784**	**0.772**	**0.93**	**13%**	**87%**	**0**	**1.76**	**1.99**	**3.76**	**0.0090**
all-MiniLM-L6-v2	0.937	0.781	0.765	0.93	13%	93%	0	1.26	2.47	3.73	0.0089

The paid model scored 0.950 against 0.937 for the free local one, a gap small enough to be noise on 15 questions.

Stage 2: Retrieval on Titans

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
semantic	0.938	0.770	0.762	1.00	13%	87%	0	0.95	2.20	3.14	0.0091
keyword (chosen)	**0.978**	**0.781**	**0.768**	**0.87**	**13%**	**87%**	**0**	**0.01**	**2.12**	**2.13**	**0.0091**
hybrid	0.944	0.784	0.779	1.00	13%	87%	0	1.00	2.39	3.39	0.0093
hybrid-rerank	0.938	0.738	0.722	0.97	13%	87%	0	1.93	2.25	4.17	0.0090

Keyword search scored highest at 0.978 but found the right paper least often, at 0.87 against 1.00 for hybrid.

Stage 2: Passages used on Titans

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
top-3	0.965	0.784	0.746	0.83	33%	67%	0	0.01	1.87	1.88	0.0060
top-5	0.978	0.781	0.768	0.87	13%	87%	0	0.01	2.12	2.13	0.0091
top-8 (chosen)	**0.972**	**0.786**	**0.774**	**0.93**	**13%**	**87%**	**0**	**0.01**	**2.20**	**2.21**	**0.0117**

Using 3 passages raised refusals to 33%, and 8 won a close tie with 5 because its relevancy was higher.

Stage 3: Prompts on TTT RNNs

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
p0-original	0.948	0.744	0.730	1.00	7%	93%	0	0.01	2.29	2.29	0.0113
p1-strict-citations (chosen)	**0.982**	**0.715**	**0.715**	**1.00**	**0%**	**100%**	**0**	**0.01**	**2.37**	**2.38**	**0.0117**
p2-step-by-step	0.963	0.710	0.710	1.00	0%	100%	0	0.01	3.03	3.04	0.0118
p3-short-extractive	0.990	0.697	0.697	1.00	0%	100%	0	0.01	1.72	1.73	0.0109
p4-by-paper	0.935	0.657	0.657	1.00	0%	100%	0	0.01	3.80	3.80	0.0155

P3 scored highest at 0.990, but P1 (0.982) won the close tie on relevancy, and the by-paper prompt P4 scored lowest.

Stage 4: Models on Hierarchical Reasoning

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
gpt-4o-mini (chosen)	**0.974**	**0.729**	**0.734**	**0.93**	**13%**	**93%**	**0**	**0.01**	**2.56**	**2.56**	**0.0127**
gpt-5-nano	0.963	0.686	0.686	0.93	0%	100%	0	0.01	3.07	3.07	0.0117
gpt-oss-20b	0.922	0.681	0.673	0.93	7%	93%	0	0.01	10.06	10.07	0.0101
gemini-3.5-flash-lite	0.967	0.688	0.697	0.93	13%	93%	0	0.01	12.45	12.46	0.0076

`gpt-4o-mini` won at 0.974, and the free Gemini model was close at 0.967 but slower. gpt-5-nano rejected the temperature setting, so it ran at its default.

Stage 4: Temperature on Hierarchical Reasoning

Value tried	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
t-0.0 (chosen)	**0.986**	**0.715**	**0.720**	**0.93**	**13%**	**93%**	**0**	**0.01**	**3.09**	**3.10**	**0.0124**
t-0.1	0.974	0.729	0.734	0.93	13%	93%	0	0.01	2.56	2.56	0.0127
t-0.4	0.946	0.749	0.744	0.93	7%	93%	0	0.01	2.87	2.87	0.0125
t-0.7	0.970	0.740	0.734	0.93	7%	93%	0	0.01	2.56	2.56	0.0131
t-1.0	0.957	0.735	0.739	0.93	13%	93%	0	0.01	3.01	3.02	0.0128

Temperature 0.0 scored highest at 0.986, but scores did not fall smoothly as it rose, because 0.7 beat 0.4, which suggests noise on 15 questions.

4 Final results on held-back questions

The five setups

Setup	Chunking	Embed.	Retrieval	k	Generator	Prompt	Temp.	Judge (prompt)
C (top score)	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
D (served)	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
E	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
A	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
B	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)

A combined every stage winner, B is the earlier project's setup, and C changed only A's generator. D and E were added after the first results, once keyword search was seen to bypass Pinecone.

Combined scores on all 60 held-back questions

Setup	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
C (top score)	0.985	0.717	0.717	0.98	10%	92%	0	0.01	10.11	10.12	0.0268
D (served)	**0.978**	**0.705**	**0.702**	**0.97**	**5%**	**97%**	**0**	**1.13**	**9.20**	**10.33**	**0.0291**
E	0.962	0.738	0.736	0.97	7%	93%	0	1.33	2.57	3.89	0.0495
A	0.956	0.738	0.734	0.98	7%	93%	0	0.01	2.23	2.24	0.0487
B	0.928	0.771	0.755	0.96	18%	82%	0	1.60	2.02	3.62	0.0252

D is 0.007 behind C on faithfulness, refuses half as often, and is 0.049 ahead of the earlier project's setup.

Faithfulness by anchor

Setup	Nested Learning	Titans	TTT RNNs	Hierarchical Reasoning
C	0.984	0.967	0.992	0.993
D	**0.979**	**0.978**	**0.979**	**0.976**
E	0.969	0.914	0.994	0.962
A	0.965	0.937	0.950	0.967
B	0.921	0.926	0.949	0.916

D ranges from 0.976 to 0.979 across anchors, the narrowest range, while C ranges from 0.967 to 0.993.

All per-anchor runs on held-back questions

Setup, anchor	Faithful.	Relev.	Rel. if ans.	Right paper	Refusals	Claims made	Judge fails	Retrieve (s)	Generate (s)	Total (s)	Cost ($)
C, Nested Learning	0.984	0.723	0.719	1.00	7%	93%	0	0.01	14.09	14.11	0.0072
C, Titans	0.967	0.733	0.745	0.97	27%	80%	0	0.01	12.07	12.08	0.0059
C, TTT RNNs	0.992	0.724	0.726	1.00	7%	93%	0	0.01	10.80	10.81	0.0068
C, Hierarchical Reasoning	0.993	0.688	0.688	0.97	0%	100%	0	0.01	3.47	3.47	0.0068
D, Nested Learning	**0.979**	**0.716**	**0.705**	**1.00**	**7%**	**93%**	**0**	**1.77**	**8.23**	**10.00**	**0.0073**
D, Titans	**0.978**	**0.697**	**0.697**	**0.97**	**0%**	**100%**	**0**	**1.03**	**7.24**	**8.27**	**0.0077**
D, TTT RNNs	**0.979**	**0.702**	**0.703**	**1.00**	**7%**	**100%**	**0**	**0.78**	**11.08**	**11.86**	**0.0075**
D, Hierarchical Reasoning	**0.976**	**0.705**	**0.702**	**0.90**	**7%**	**93%**	**0**	**0.93**	**10.27**	**11.20**	**0.0066**
E, Nested Learning	0.969	0.728	0.728	1.00	0%	100%	0	1.77	2.94	4.71	0.0137
E, Titans	0.914	0.763	0.758	0.97	20%	80%	0	0.75	1.99	2.75	0.0112
E, TTT RNNs	0.994	0.745	0.745	1.00	0%	100%	0	1.70	2.67	4.36	0.0124
E, Hierarchical Reasoning	0.962	0.715	0.715	0.90	7%	93%	0	1.08	2.67	3.75	0.0121
A, Nested Learning	0.965	0.730	0.719	1.00	7%	93%	0	0.01	2.54	2.55	0.0125
A, Titans	0.937	0.757	0.757	0.97	20%	80%	0	0.01	2.12	2.13	0.0113
A, TTT RNNs	0.950	0.730	0.730	1.00	0%	100%	0	0.01	2.10	2.10	0.0123
A, Hierarchical Reasoning	0.967	0.734	0.734	0.97	0%	100%	0	0.01	2.16	2.17	0.0126
B, Nested Learning	0.921	0.759	0.744	0.97	13%	87%	0	1.48	2.08	3.56	0.0071
B, Titans	0.926	0.800	0.783	1.00	27%	73%	0	1.38	1.83	3.21	0.0059
B, TTT RNNs	0.949	0.777	0.761	0.97	13%	87%	0	2.10	1.97	4.06	0.0063
B, Hierarchical Reasoning	0.916	0.749	0.733	0.90	20%	80%	0	1.45	2.20	3.65	0.0059

These are the 20 child runs behind the combined scores. Latency varies most for the free Gemini model, which may reflect free-tier load.

5 Why setup D is served

- **It uses Pinecone:** keyword-only retrieval never queries Pinecone when answering, while D searches by keywords and by meaning, as the proposal describes.
- **The test favours keywords:** question rules told the generator to name the method being asked about, so exact-word matching has an easy job. Real users paraphrase.
- **It refuses less and is more consistent:** it refuses 5% of questions against 10% for C.
- **The trade-offs:** D's relevancy when answered is 0.702 against 0.755 for B, and it takes 10.3 seconds against 3.6. Setup E is the faster alternative, at 3.9 seconds and 0.962 faithfulness.

6 Full configuration of every run

TE3S is `text-embedding-3-small`, MiniLM is `all-MiniLM-L6-v2`, and Gemini Lite is `gemini-3.5-flash-lite`. All runs used 15 retrieval candidates and at most 512 output tokens. k is the number of passages used.

Runs 1 to 32: choosing settings

#	Stage	Setting	Value	Anchor	Chunking	Embed.	Retrieval	k	Generator	Prompt	Temp.	Judge (prompt)
1	1	judge	gpt-4o-mini	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (binary)
2	**1**	**judge**	**gpt-4o-mini-rubric**	**Nested Learning**	**recursive 400/80**	**TE3S**	**hybrid + rerank**	**5**	**gpt-4o-mini**	**p0**	**0.1**	**gpt-4o-mini (rubric)**
3	1	judge	gpt-5-nano	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-5-nano (binary)
4	1	judge	gemini-3.5-flash-lite	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	Gemini Lite (binary)
5	1	judge	gpt-oss-20b	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-oss-20b (binary)
6	1	chunking	fixed-512-50	Nested Learning	fixed 512/50	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
7	1	chunking	recursive-400-80	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
8	**1**	**chunking**	**recursive-800-120**	**Nested Learning**	**recursive 800/120**	**TE3S**	**hybrid + rerank**	**5**	**gpt-4o-mini**	**p0**	**0.1**	**gpt-4o-mini (rubric)**
9	1	chunking	semantic-p70	Nested Learning	semantic p70	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
10	**1**	**embedding**	**text-embedding-3-small**	**Nested Learning**	**recursive 800/120**	**TE3S**	**hybrid + rerank**	**5**	**gpt-4o-mini**	**p0**	**0.1**	**gpt-4o-mini (rubric)**
11	1	embedding	all-MiniLM-L6-v2	Nested Learning	recursive 800/120	MiniLM	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
12	2	retrieval mode	semantic	Titans	recursive 800/120	TE3S	semantic	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
13	**2**	**retrieval mode**	**keyword**	**Titans**	**recursive 800/120**	**TE3S**	**keyword**	**5**	**gpt-4o-mini**	**p0**	**0.1**	**gpt-4o-mini (rubric)**
14	2	retrieval mode	hybrid	Titans	recursive 800/120	TE3S	hybrid	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
15	2	retrieval mode	hybrid-rerank	Titans	recursive 800/120	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
16	2	passages used	top-3	Titans	recursive 800/120	TE3S	keyword	3	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
17	2	passages used	top-5	Titans	recursive 800/120	TE3S	keyword	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
18	**2**	**passages used**	**top-8**	**Titans**	**recursive 800/120**	**TE3S**	**keyword**	**8**	**gpt-4o-mini**	**p0**	**0.1**	**gpt-4o-mini (rubric)**
19	3	prompt	p0-original	TTT RNNs	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
20	**3**	**prompt**	**p1-strict-citations**	**TTT RNNs**	**recursive 800/120**	**TE3S**	**keyword**	**8**	**gpt-4o-mini**	**p1**	**0.1**	**gpt-4o-mini (rubric)**
21	3	prompt	p2-step-by-step	TTT RNNs	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p2	0.1	gpt-4o-mini (rubric)
22	3	prompt	p3-short-extractive	TTT RNNs	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p3	0.1	gpt-4o-mini (rubric)
23	3	prompt	p4-by-paper	TTT RNNs	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p4	0.1	gpt-4o-mini (rubric)
24	**4**	**model**	**gpt-4o-mini**	**Hierarchical Reasoning**	**recursive 800/120**	**TE3S**	**keyword**	**8**	**gpt-4o-mini**	**p1**	**0.1**	**gpt-4o-mini (rubric)**
25	4	model	gpt-5-nano	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-5-nano	p1	0.1	gpt-4o-mini (rubric)
26	4	model	gpt-oss-20b	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-oss-20b	p1	0.1	gpt-4o-mini (rubric)
27	4	model	gemini-3.5-flash-lite	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.1	gpt-4o-mini (rubric)
28	**4**	**temperature**	**t-0.0**	**Hierarchical Reasoning**	**recursive 800/120**	**TE3S**	**keyword**	**8**	**gpt-4o-mini**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
29	4	temperature	t-0.1	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.1	gpt-4o-mini (rubric)
30	4	temperature	t-0.4	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.4	gpt-4o-mini (rubric)
31	4	temperature	t-0.7	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.7	gpt-4o-mini (rubric)
32	4	temperature	t-1.0	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	1.0	gpt-4o-mini (rubric)

Each group changes one setting, and chosen rows carry forward. Some rows share saved answers when their settings match, so they cost nothing extra to run.

Runs 33 to 57: held-back questions

#	Stage	Setting	Value	Anchor	Chunking	Embed.	Retrieval	k	Generator	Prompt	Temp.	Judge (prompt)
33	Final	setup	C-best-free-generator	All 4 (parent run)	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
34	Final	setup	C-best-free-generator	Nested Learning	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
35	Final	setup	C-best-free-generator	Titans	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
36	Final	setup	C-best-free-generator	TTT RNNs	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
37	Final	setup	C-best-free-generator	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	Gemini Lite	p1	0.0	gpt-4o-mini (rubric)
38	**Final**	**setup**	**D-hybrid-free-generator**	**All 4 (parent run)**	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
39	**Final**	**setup**	**D-hybrid-free-generator**	**Nested Learning**	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
40	**Final**	**setup**	**D-hybrid-free-generator**	**Titans**	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
41	**Final**	**setup**	**D-hybrid-free-generator**	**TTT RNNs**	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
42	**Final**	**setup**	**D-hybrid-free-generator**	**Hierarchical Reasoning**	**recursive 800/120**	**TE3S**	**hybrid**	**8**	**Gemini Lite**	**p1**	**0.0**	**gpt-4o-mini (rubric)**
43	Final	setup	E-hybrid-gpt-4o-mini	All 4 (parent run)	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
44	Final	setup	E-hybrid-gpt-4o-mini	Nested Learning	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
45	Final	setup	E-hybrid-gpt-4o-mini	Titans	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
46	Final	setup	E-hybrid-gpt-4o-mini	TTT RNNs	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
47	Final	setup	E-hybrid-gpt-4o-mini	Hierarchical Reasoning	recursive 800/120	TE3S	hybrid	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
48	Final	setup	A-all-winners	All 4 (parent run)	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
49	Final	setup	A-all-winners	Nested Learning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
50	Final	setup	A-all-winners	Titans	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
51	Final	setup	A-all-winners	TTT RNNs	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
52	Final	setup	A-all-winners	Hierarchical Reasoning	recursive 800/120	TE3S	keyword	8	gpt-4o-mini	p1	0.0	gpt-4o-mini (rubric)
53	Final	setup	B-control-baseline	All 4 (parent run)	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
54	Final	setup	B-control-baseline	Nested Learning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
55	Final	setup	B-control-baseline	Titans	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
56	Final	setup	B-control-baseline	TTT RNNs	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)
57	Final	setup	B-control-baseline	Hierarchical Reasoning	recursive 400/80	TE3S	hybrid + rerank	5	gpt-4o-mini	p0	0.1	gpt-4o-mini (rubric)

Each setup has one parent run and 4 per-anchor child runs. Together with runs 1 to 32, this covers all 57 MLflow runs.

7 Limits of these results

- **Small samples:** each stage used 15 questions on one anchor, so small gaps may be noise, and a winner on one anchor may not win elsewhere.
- **Stage winners do not simply add up:** changing only A's generator gave C a higher score than the combined winners.
- **Questions were written by an LLM:** they test finding and using what is in the corpus, not what real researchers would ask. The team removed 13 that did not stand alone.
- **Scores are LLM judgements:** they vary slightly between runs, and `gpt-4o-mini` judges its own answers in setups A, B and E.
- **Faithfulness has blind spots:** an answer can be faithful to the wrong passages, so the right-paper rate is shown alongside it.
- **Latency is not production latency:** it was measured on a laptop CPU, with free-tier models.

- **The history was rebuilt:** MLflow runs were re-logged from saved answers after fixing measurement bugs, so MLflow's own timestamps and durations reflect that step, not the original model calls.
- **Added setups:** D and E were tested after seeing A, B and C, so their comparison is slightly less clean than a fully planned one.